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Low Thermal Budget Growth of Near-Isotropic Diamond Grains for Heat Spreading in Semiconductor Devices

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The ever-increasing power density is a major trend for electronics applications from dense computing to 5G/6G networks. Joule heating and resulting high temperature in the device channel due to the increased power density results in performance degradation and premature failure. Diamond integration near the hot spot can spread the heat by increasing the heat transfer coefficient. Diamond is mostly grown at high temperatures (700–1000 °C), which limits its integration with many semiconductor technologies. Here, a high-quality 400 °C-diamond by modifying the gas chemistry at different nucleation stages, with a sharp sp^3 Raman peak (FWHM≈6.5 cm^{-1}) and high phase purity (97.1%), similar to 700 °C-diamond (>98%) is demonstrated. An average grain size of 650 nm with a thickness of 790 nm corresponding to an anisotropy ratio of 1.21 at 400 °C close to the best-reported of 1.12 at 700 °C is achieved. This near-isotropic diamond exhibits a relatively high thermal conductivity of ≈300 W m^{-1} K^{-1} and a thermal boundary resistance as small as only 5 m^2K G^{-1} W^{-1} (on SiO_2 and Si_3N_4). Achieving such a high-quality diamond at 400 °C demonstrates the possibility to grow the diamond on a wide range of semiconductors including Si, InP, Ga_2O_3, SiC, and GaN where SiO_2 or its variations, and Si_3N_4 are commonly used dielectrics.

1. Introduction

With ultra-scaled semiconductor devices, self-heating is a common problem that makes thermal management of electronics at every possible level essential for sustaining higher power densities in almost all applications, including computation, 5G/6G, and power electronics.[1–4] Self-heating under operating conditions leads to increased junction temperatures. The elevated channel temperature degrades device performance, by reducing channel mobility, which further increases the self-heating due to increased channel resistance, thus setting up a cycle to cause premature failure.[5,6] Diamond is an excellent material to aid in extracting heat from these devices to reduce channel temperature.[7,8] Our prior work has demonstrated polycrystalline (PC) diamond on GaN with grains larger than 2 µm offering a thermal conductivity (TC) > 650 W m^{-1} K^{-1}, which is larger than that of Si, SiC, GaN, and any dielectric material's TC. Such high-quality diamond grains can be used as a heat-spreading layer if it is integrated close to the hot spot.[9–11] 2D materials such as graphene and h-BN have been considered for heat spreading as well.[12–14] Diamond has sp^3 hybridized carbon atoms that attach to four other carbon atoms located at the vertices of a tetrahedron that makes a continuous and uniform 3D network of C—C sigma bonds, thus it has superior cross-plane TC to 2D materials. This perfect 3D periodicity and large Debye temperature make diamond a superior heat conductor compared to 2D materials, semiconductors, and even metals when heat conduction in all directions is desired.[15] To maximize the thermal management capabilities of PC diamond in 3D, an isotropic poly-grained layer with similar in-plane and cross-plane TCs is preferred.[16] Isotropic PC diamond grains lessen the need for thick diamond layers for higher TC (unlike conventional growth[17]) and lowers the thermal residual stress in the diamond layer after the growth due to coefficient of thermal expansion (CTE) mismatch.[18]

Diamond growth at its typical temperatures of 650–900 °C imposes challenges for semiconductors processes that require low thermal budget diamond integration. Extensive damages to the substrate or device channel performance (e.g., mobility and carrier concentration) can occur while growing the diamond near the hot spot in the channel. The ability to grow diamond at a reduced growth temperature would offer unparalleled advantages, for example, diamond could be overgrown on metal contacts without degrading them. Low-temperature diamond can be integrated at different stages of the CMOS processing as well (such as back-end-of-line (BEoL)) which is only compatible with temperatures below 450 °C.[19,20] However, lowering the temperature to less than 600 °C during diamond growth results in higher sp^2 carbon incorporation which reduces sp^3 (diamond) phase purity and deteriorates thermal properties (e.g., TC) of the diamond layer. Thus, oxygen species (like O_2)

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were introduced to the chamber and optimized to enhance sp^2 etching at lower temperatures and to promote sp^3 formation. The added oxygen during low temperature growth takes the role of high temperature H_2-plasma during high-temperature diamond growth by increasing the dissociation rate of methane monomers and diamond nucleation and accelerating the non-diamond etching rate.[21] Low-temperature diamond growth has been reported in a wide temperature range from 100 to 500 °C in various gas precursors combinations such as CO/H_2, $CO/O_2/H_2$,[22] $CO_2/CH_4/H_2$,[23] Ar/CH_4,[24,25] $Ar/CH_4/H_2$,[26] and $O_2/CH_4/H_2$.[21,27,28] In contrast to previous studies on low-temperature grown diamonds to date, our goal is to not only reduce the growth temperature to <400 °C but to also keep the thermal properties, phase purity, and anisotropy ratio similar to that of > 600 °C grown diamond. This near-isotropic, low-temperature-grown diamond has been developed on SiO_2 and Si_3N_4 thin layers and can be integrated on all semiconductor-based devices with SiO_2 or Si_3N_4 dielectric capping.

2. Results and Discussions

2.1. Diamond on SiO_2

2.1.1. CH_4/H_2 Gas System

In our study, it was first determined that using only CH_4 and H_2 gases, which is typically used in standard high temperature (700 °C) diamond growth had limitations (in blocking sp^2 carbons) at lower temperatures. In this section, the structural properties of diamonds grown at 300 to 700 °C with CH_4/H_2 gases are compared. As shown in **Figure 1**, by reducing the temperature, the grain shape, size, and crystallites change drastically. 700 °C-grown diamond shows the largest grains and 300 °C-grown diamond the smallest grains. In CH_4/H_2 gas system an activation energy of 20–30 kcal mol^{-1} is required for the abstraction of hydrogen atoms from C—H bonds at the surface to replace it with another carbon atom. At lower temperatures fulfilling this activation energy is challenging.[25] Lower temperatures H_2-plasma is less efficient at etching sp^2 carbons and promoting sp^3 formation over sp^2, thus the growth resulted in a soot-like carbon with significantly smaller grain sizes. This soot-like carbon layer was delaminated from the substrate easily due to the lack of strong adhesion and high residual stress. The 400 °C-grown diamond shows increased grain sizes and more facet-like diamond grains compared to 300 °C diamond due to the 100 °C increase in temperature. However, sp^2 carbon is still incorporated; consequently, the re-nucleation rate is high which causes new diamond grains to form on top of each other instead of enlarging the existing grains. The crystal quality and phase purity were compared using Raman spectroscopy (Figure 1). As a reference, the Raman spectra of a single crystalline diamond anvil cell was assessed to compare with polycrystalline samples. The Raman peak position and FWHM for the single crystalline diamond were measured as 1331.5 and 4.9 cm^{-1}, respectively.[9]As expected, 700 °C-grown diamond exhibited higher crystalline quality (FWHM of sp^3 peak ≈5.7 cm^{-1}[9]) and higher phase purity (98%[18]) compared to lower temperature grown polycrystalline diamonds. By decreasing the temperature, the FWHM increases (7.94 cm^{-1} at 500 °C, 9.27 cm^{-1} at 400 °C, 21.36 cm^{-1} at 300 °C) and the phase purity deteriorates (94.9% at 500°C, 84.2% at 400°C, 75.4% at 300 °C) due to the incorporation of sp^2 carbon. As a result, a different gas system is necessary to grow high quality diamonds at temperatures < 500 °C.

2.1.2. $O_2/CH_4/H_2$ Gas System

The poor-quality diamond grown at low temperature using a CH_4/H_2 gas system described in the previous section motivated the study of a different gas chemistry that will not only enhance sp^2 (disordered) carbon etching but also lower the sp^3 carbon growth activation energy. To enable a diamond growth at 400 °C, that would show similar structural properties as the 700 °C-grown diamond, the addition of oxygen to the gas mixture and nucleation optimization were two necessary steps, described in this section. As shown in **Figure 2**a, adding oxygen to the chamber etches carbon due to the high electronegativity of oxygen atoms. Oxygen preferentially attracts the electrons from the surface carbon-carbon bonds as carbon is not nearly as electronegative, hence breaking the carbon-to-carbon bonds.

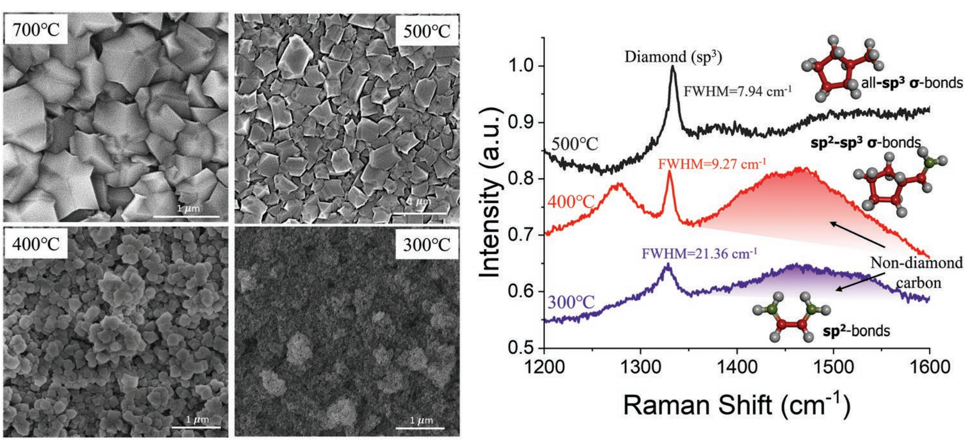

Figure 1. Plan-view scanning electron microscopy (SEM) micrographs and Raman spectra of diamonds were grown at different temperatures in only H_2/CH_4 gas environment. Diamond structural quality and phase purity was degraded by reducing temperature.

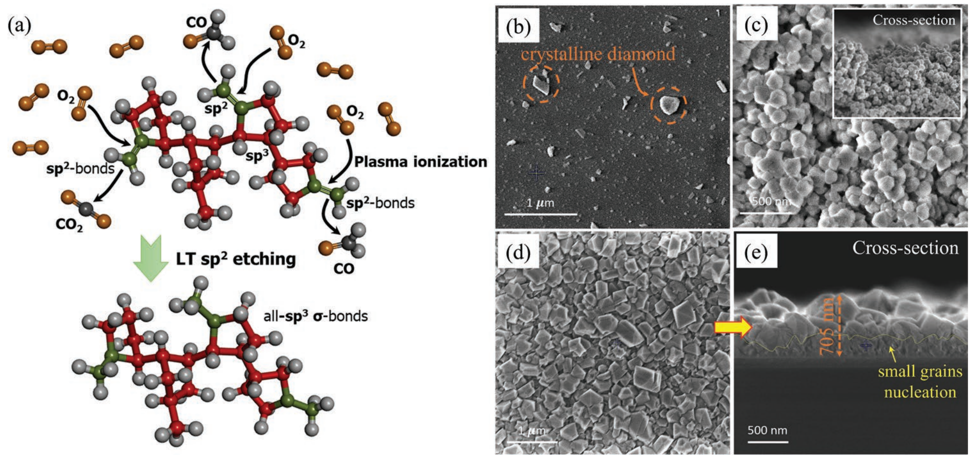

Figure 2. a) Oxygen plasma impact on carbon bonding by etching sp^2 carbon double bonds. The more reactive nature of sp^2 double bonds over sp^3 single bonds results in sp^2 carbon bonds being preferentially etched by atomic oxygen. b)–e) SEM micrographs of diamonds were grown at 400 °C with various nucleation and growth parameters.

Between sp^3 and sp^2 bonds sp^2 carbon bonds are preferentially etched because while carbon-carbon double bonds are overall stronger, the individual bond energies of the two bonds are lower on average and so are more reactive than the carbon-carbon single bond, especially since the single bonds are mostly in rings which are extremely stable.

On sample #B_1 a 3.4% CH_4 and relatively high O_2 content (2%) was used for diamond growth. As shown in Figure 2b, the grown layer is nonuniform with many diamond islands which did not coalesce. Higher etch rate of carbon due to the initiation of oxygen plasma and extremely low growth rate due to the insufficient atomic carbons (CH_3^+) reduced the nucleation rate noticeably. The lack of appropriate diamond nucleation during the earlier stages of the growth resulted in the formation of sparse diamond particles. Even though, oxygen plasma enhances the dissociation process of carbon atoms from CH_4 for higher growth rate, it etches carbon-carbon bonds simultaneously. Furthermore, exposing the regions of SiO_2, uncovered by the diamond particles, to O_2-plasma even at lower temperatures increases the chance of roughening or etching of the surface which was observed in sample #B_1. For sample #B_2, a 1% O_2 and a 5% CH_4% were used as shown in Figure 2c, higher re-nucleation rate than in sample #B_1 resulted in a uniform diamond coverage of the substrate but with much smaller grains. The inset cross-section SEM verifies high re-nucleation rate; however, ultra-nano-crystalline diamond (UNCD) growth was observed instead of desired micro-crystalline diamond (MCD) growth. To effectively nucleate and simultaneously decrease the re-nucleation rate to form larger grains, a combination of sample #B_1 and #B_2 growth parameters were used to achieve the results shown in Figure 2d,e for sample #B_3. Using a relatively higher O_2 concentration right after the high re-nucleation stage makes the etching and growth comparable which reduces sp^2 carbon formation and at the same time minimizes the possibility of substrate damage (5–3%CH_4, 1–2%O_2).

When the lateral and vertical growth rates are comparable isotropic grains of diamond is achievable. Isotropic grains provide higher effective TC, eliminating the need for thick diamond layers. The anisotropy ratio of the crystal grains for sample #B_3 was ≈1.76 (705/400 nm) which is close to our goal of unity that was achieved in 700 °C growth. However, a thick nucleation layer with smaller grains (Figure 2e) negatively affected the quality and phase purity of this layer which eventually can degrade its thermal properties. As extracted from Raman spectra in **Figure 3**a, the FWHM of samples #B_2 and #B_3 were 11.5 and 9.3 cm^{-1}, respectively. This large FWHM along with high sp^2 incorporation lowers their phase purities to 83.6% and 92.5%. Thus, to increase phase purity, a 3-step growth technique was developed to simultaneously reduce sp^2 carbon and grow a uniform and completely coalesced layer as shown in Figure 3b (5%CH_4,1.7% O_2→3.4%CH_4,1.7%O_2→2.6%CH_4,2%O_2). With this new technique, not only was the sp^2 incorporation decreased (blue data in Figure 3a), but also the small grain nucleation layer has disappeared, as shown in Figure 3c. Additionally, the FWHM was reduced to 8.8 cm^{-1}, another sign of higher quality diamond than samples #B_2 and #B_3. For sample #B_4, the anisotropy ratio was dropped to 1.21, significantly closer to 1.12 for 700 °C-grown diamond (inset Figure 3c).

Due to the mismatch in the coefficients of thermal expansion (CTE) between the diamond and the substrate (GaN/SiC), a residual stress was present in the diamond layer. Since the diamond's CTE (1.1×10^{-6} K^{-1}) is lower than the substrate, the stress after cooling from 400 °C deposition is compressive. In sample #B_2, a red shift of 2.6 cm^{-1} from sp^3 (diamond) peak position was measured, which is a sign of tensile stress in opposition to our expectations due to a CTE mismatch. The main reason for this observation is the high re-nucleation rate which relaxes the diamond layer by UNCD formation. This layer as confirmed by SEM (Figure 2c) has more grain boundaries and polyacetylene which eventually results in tensile stress by compensating the compressive stress caused by CTE mismatch.[29] On the other hand, Samples #B_3 and #B_4 have a blue shift of 1.3 and 2.1 cm^{-1}, respectively. Since their thicknesses are in the same range, the higher compressive stress in sample #B_4 is due to the lower re-nucleation rate and sp^2 incorporation.

To reduce the FWHM of the diamond was grown at 400 °C (to make it closer to 700 °C diamond), we have optimized the first step, the nucleation stage, by sweeping the CH_4%. As can

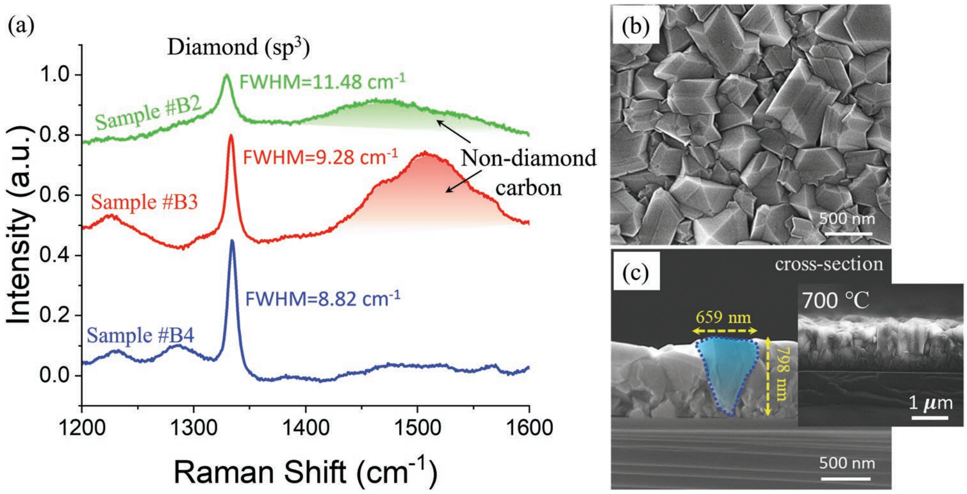

Figure 3. Raman spectra and SEM micrographs of diamonds were grown at 400 °C with various nucleation and growth parameters.

be seen in the Raman spectroscopy results in **Figure 4**, the lowest CH_4% (3.4%) during the first step nucleation that enables a uniform layer growth resulted in a narrower sp^3 peak with a smaller 6.6 cm^{-1} FWHM. While the top-view SEMs confirm no change in the grain size, grain shapes, and uniformity, lower CH_4% nucleation resulted in a narrower diamond peak and improved crystallinity.

All the key results presented above including FWHM, phase purity, Raman shift corresponding to residual stress, and anisotropy ratio are summarized in **Table 1** for comparison.

2.1.3. Thermal and Interface Analysis

In order to increase the heat transport coefficient of diamond heat spreader its TC must be as high as possible and the thermal boundary resistance (TBR) between diamond and the substrate needs to be as low as possible. Our optimized low-temperature diamond (sample #B_4 in Figure 3c) has a grain size of 659 nm which in theory corresponds to an average TC between 300 and 400 W m^{-1} K^{-1}.[30] Using the transient thermoreflectance (TTR) method and fitting the analytical model to the measured signal (**Figure 5**), we have measured TC and TBR of the diamond layer. The 400 °C-grown diamond TC was measured in the range of 250 to 400 W m^{-1} K^{-1} with a mean value of ≈300 W m^{-1} K^{-1}, consistent with the similar grain sizes grown at 700 °C. The measured TC for our 400 °C-diamond is ≈200% higher than the TC was reported by[26] (≈110 W m^{-1} K^{-1} measured at 20 °C) for 300 nm-thick NCD diamond film which was grown at 450–500 °C. This relatively high TC with only 798 nm of thickness is dedicated to the near-isotropic shape of the grains which decreases the phonon scattering rate in these

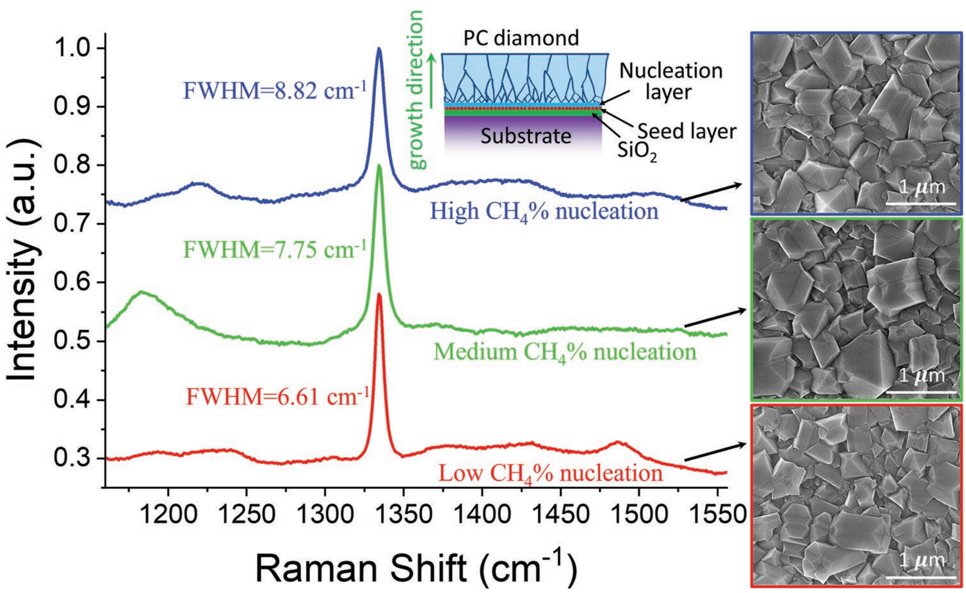

Figure 4. Diamond nucleation optimization at the first step for 3-steps technique by sweeping the CH_4%, which has reduced the FWHM to 6.6 cm^{-1}.

Table 1. A summary of key results for different samples were studied.

Sample#	Phase Purity^a) [%]	FWHM [cm^{-1}]	Raman shift^b) [cm^{-1}]	Residual Stress^e) [GPa]	Anisotropy ratio^c)
A_1	> 98[10,18]	5.6[9]	+2.7[10]	−1.53	1.12[9]
A_2	94.9	7.9	+0.5	−0.28	≈2.53
A_3	82.4	9.3	−2.3	+1.30	–d)
A_4	75.4	21.4	−3.3	+1.87	–d)
B_2	83.6	11.5	−2.6	+1.47	–d)
B_3	92.5	9.3	+1.3	−0.73	≈1.76
B_4	97.1	8.8	+2.1	−1.19	≈1.21
B_5	Similar to B_4	7.8	+2.2	−1.24	≈1.21
B_6	Similar to B_4	6.6	+2.3	−1.30	≈1.21

a)The phase purity has been calculated using the method explained in[18]; b)Higher re-nucleation rate applies tensile stress to the diamond layer and compensates for part of the compressive stress due to the CTE mismatch. These Raman shift values were measured on diamond grown on GaN/SiC substrates; c)Anisotropy ratio is diamond thickness divided by its grain size; d)Too large to be calculated; e)The residual stress has been calculated using the method explained in[18,36,37] and the Raman peak position shift.

grains and eventually enhances its thermal properties. Another advantage of this low temperature diamond is the high-quality nucleation at the interface which resulted in a TBR as low as 5 $m^2K\ G^{-1}\ W^{-1}$. This TBR is extremely close to 3.1 $m^2K\ G^{-1}\ W^{-1}$ which was reported in one of our previous publications for high temperature diamond grown at 700 °C.[9] Figure 5b shows EELS mapping of carbon atoms with the change in the chemical bonding at the interface. An extremely sharp transition from SiO_2 to sp^3 carbon (point 4⟶3) can be observed confirming a thin interfacial layer (less temperature profile discontinuity) and high-quality nucleation which supports our low TBR.

2.2. Diamond on Si_3N_4

In order to extend low temperature diamond application, we have utilized our optimized low temperature (400 °C) diamond growth technique from Section 2.1.2 on Si_3N_4 to study the interface smoothness and diamond crystallite. As shown in **Figure 6**a the shape of the grains as well as the diamond nucleation is extremely similar to SiO_2 substrates which emphasize the broadness of our seeding/nucleation method. We have achieved a smooth interface with an average grain size between 500 and 700 nm. The interface between diamond and Si_3N_4 has been characterized using energy-dispersive spectroscopy (EDS) as shown in Figure 6b confirming a thin transition from carbon to Si_3N_4 which can enhance its thermal properties by reducing the thermal barrier thickness.

3. Conclusion

We have shown that low temperature high quality diamond can be grown on top of SiO_2 or Si_3N_4. With the low temperature growth of diamond on these two widely used dielectrics in various semiconductor technologies such as Si, GaN, SiC, InP, and Ga_2O_3, this work opens up a path for heat spreading solutions for wider applications. Reducing diamond growth temperature not only can expand the growth window for diamond on GaN-based devices but also enable the integration of diamond on other semiconductor devices as a passivation layer or inter-layer dielectric. The use of only hydrogen plasma, typically

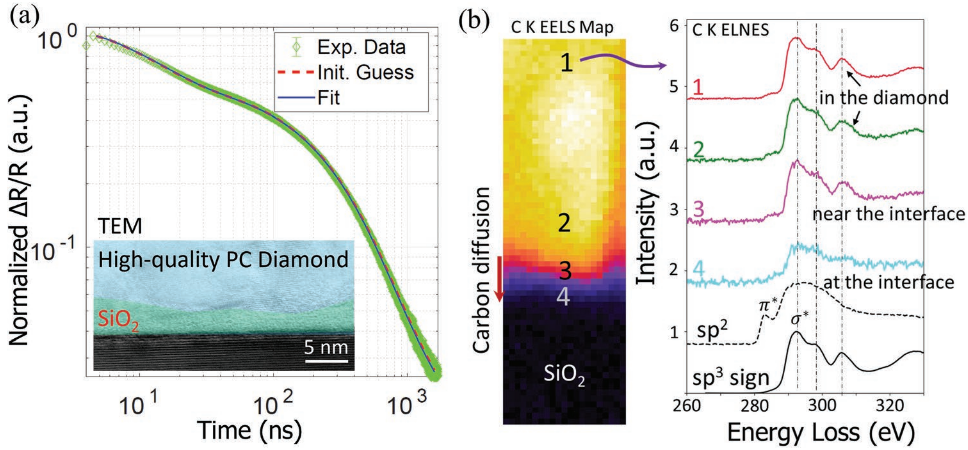

Figure 5. a) Normalized thermoreflectance signal as a function of time; lines represent the experimental value and dots represent an analytical model fitted to the experimental values. b) EELS analysis at the diamond-SiO_2 interface showing chemical bonding type at various points.

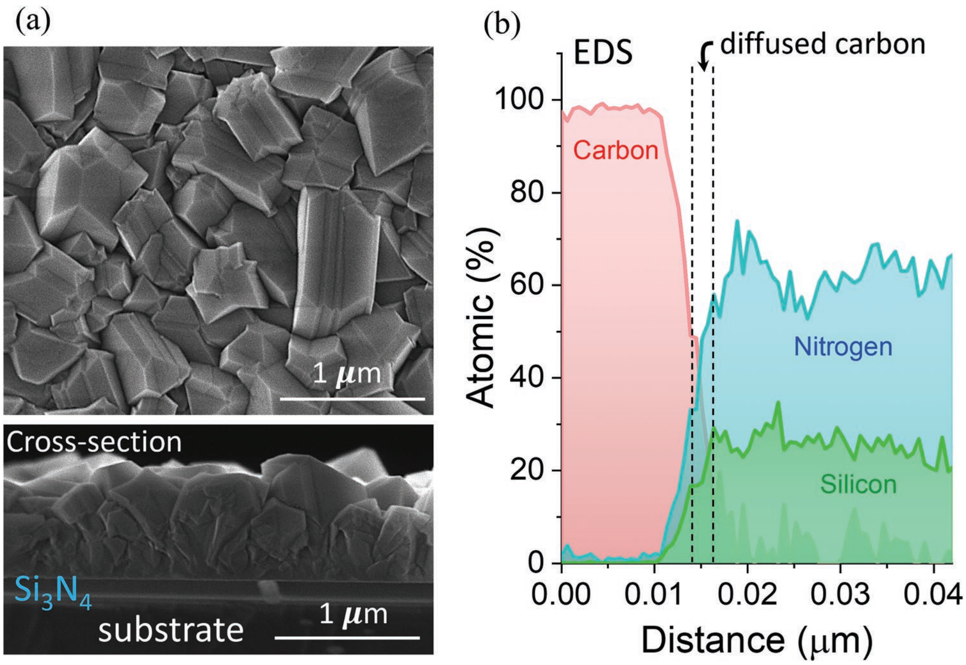

Figure 6. a) Plan-view and cross -sectional view SEM micrographs of diamonds were grown at 400 °C on Si_3N_4. b) EDS results of PC diamond grown on Si_3N_4.

used in diamond high temperature growth, during lower temperature growth was found to be insufficient. This resulted in deteriorated diamond film quality due to a reduced etch rate of non-diamond carbons and an increased incorporation of sp^2 carbon. Here, we have developed a low temperature 3-step diamond growth technique at 400°C by optimizing the gas mixture ($H_2/CH_4/O_2$), which produced diamond growth with similar quality, phase purity, and morphology to that of 700 °C-grown diamond. Among all the efforts in low temperature diamond growth, we have achieved isotropic grains at 400 °C with the lowest known anisotropy ratio of 1.21. This value is close to 1.12, the anisotropy ratio of our previously 700 °C grown diamond. The low anisotropy ratio provides a higher in-plane thermal conductivity as a plus to its significantly large cross-plane thermal conductivity which is required for extra efficient device cooling in all directions. We have measured an average thermal conductivity of ≈300 W m^{-1} K^{-1} and a thermal boundary resistance as low as 5 m^2K $G^{-1}W^{-1}$, which is higher than most thermal materials in their polycrystalline form.

4. Experimental Section

Prior to diamond growth, all the samples were seeded by diamond particles using a mixed ultrasonication-polymer-assisted seeding technique. Then, PC diamond thin films were grown in an SDS 5000 series microwave plasma CVD reactor from Seki Diamond Systems.[18,31] The reactant gases consisted of CH_4, the precursor to diamond, H_2, the reaction activator for C—C bonding by providing H-free reaction sites, and O_2 to minimize sp^2 carbon formation during low temperature growth. Multiple parameters were studied to optimize low temperature growth, which is summarized for each sample in **Table 2**. Plasma powers of 900–1300 W, chamber pressures of 20–40 Torr, temperatures of 300–500 °C, CH_4 content of 0.5–5%, and O_2 content of 1–2% were employed.

Table 2. Growth parameters for different samples were utilized for this study.

Sample#	Temperature [°C]	Power [W]	Pressure [Torr]	Gas System	Details
A_1	700	1600	60	H_2/CH_4	HT stage[a)]
A_2	500	1200	40	H_2/CH_4	HT stage
A_3	400	1300	45	H_2/CH_4	LT stage[b)]
A_4	300	950	30	H_2/CH_4	LT stage
B_1	400[c)]	1300	45	$H_2/O_2/CH_4$	Low CH_4% High O_2%[d)]
B_2	400	1300	45	$H_2/O_2/CH_4$	High CH_4% Low O_2%
B_3	400	1300	45	$H_2/O_2/CH_4$	2-steps technique
B_4	400	1300	45	$H_2/O_2/CH_4$	3-steps technique
B_5	400	1300	45	$H_2/O_2/CH_4$	3-steps, Medium CH_4%
B_6	400	1300	45	$H_2/O_2/CH_4$	Optimized nucleation

[a)]HT stage: High temperature stage; [b)]LT stage: Low Temperature stage; [c)]All of #B sample series were grown at 400°C on LT stage; [d)]More details in "Results and Discussion" section.

Table 3. Fixed thermal properties used for TTR fitting. The TC of layers other than diamond were extracted from control samples as indicated with *.

	TC [W/m/K]	Heat Capacity J/kg/K]	Density [kg m^{-3}]
Au	315	129	19300
$TBR_{Cr\text{-}Diamond}$	0.3 (10 nm)*	–	–
Diamond	Fitted	520	3510
$TBR_{Diamond\text{-}SiO2}$	Fitted	–	–
SiO_2	1.18*	600	2690
Si/GaN substrate[a)]	105/150*	700/470	2329/6150

[a)]2 different substrates were used in this study under the SiO_2 thin layer.

After the diamond growth, the diamond layer's grain size and thickness were determined by scanning electron microscopy (FEI Nova NanoSEM 430). Scanning transmission electron microscopy (STEM), electron energy loss spectroscopy (EELS), and energy-dispersive spectroscopy (EDS) analysis (JEOL JEM 2100F-AC) were used for interface study and identification of the chemical bonding types at the interface. To assess the diamond crystalline quality, Raman spectroscopy measurements were made (HORIBA Scientific LabRAM HR Evolution spectrometer). The full-width-half-maximum (FWHM), relative areas under sp^3 and sp^2 peaks, and red/blue shifts of Raman peaks were used to characterize the diamond quality, sp^3 phase purity, and residual thermal stress, respectively. These properties were extracted from the Raman spectra using the method explained elsewhere.[18,29,32,33] TTR was utilized for measuring the thermal properties of diamond including its TC and TBR with the substrate. Measurements were carried out on gold coated (as a transducer) diamond samples using a CW 532 nm probe laser along with a 355 nm passively Q-switched Nd:YAG pump laser with 1 ns pulse width, 10 kHz repetition rate, and $1/e^2$ spot size of ≈90 µm. Since multilayers of different materials are involved, the thermoreflectance data was fitted to the transient heat equation using the transmission-line-axis-symmetric model to extract both TBR and TC.[34,35] The details of fixed and extracted parameters from the control samples without diamonds are shown in **Table 3**.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

heat spreaders, low-temperature diamonds, self-heating, semiconductor devices, thermal management

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